INCH-POUND

MIL-PRF-19500/666
3 August 1999

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED
(TOTAL DOSE AND SINGLE EVENT EFFECTS) TRANSISTOR, P-CHANNEL
SILICON TYPE 2N7454U1, 2N7455U1
JANSD, R

This specification is approved for use by all Departments and
Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a P-Channel, enhancement-mode, MOSFET, radiation hardened (Total Dose and Single Event characterization - see figure 4), power transistor. One level of product assurance is provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1, SMD-2 (Surface mount).

1.3 Maximum ratings. T_A = +25°C, unless otherwise specified.

Type	P_T 1/ T_C = +25°C	V_{DS}	V_{DG}	V_{GS}	I_{D1} 2/ T_C =+25°C	I_{D2} T_C = +100°C	I_S 2/	I_{DM}	T_J and T_{STG}	V_{ISO} 70,000 ft. altitude
	W	V dc	V dc	V dc	A dc	A dc	A dc	A (pk)	°C	V dc
2N7454U1	208	-100	-100	±20	47.0	30.0	47.0	141	-55 to +150	N/A
2N7455U1		-200	-200		28.0	18.0	28.0	84		N/A

1/ Derate linearly 1.67 W/°C for T_C > +25°C; $P_T = (T_{jmax} - T_C)/R_{\theta JC}$

2/ $I_D = ((T_{jmax} - T_C)/((R_{\theta JC})X (r_{DS(on)} \text{ at } T_{jmax})))^{1/2}$

1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0V I_D = 1.0 mA dc	$V_{GS(TH)1}$ $V_{DS} = V_{GS}$ I_D = 1.0 mA dc		Max I_{DSS1} V_{GS} = 0 V_{GS} = 80% of rated V_{DS}	Max $r_{DS(on)}$ 1/ V_{GS} = -12V		$R_{\theta JC}$ Max	$I_{AS} = I_{DM}$
					T_J = 25°C at I_{D2}	T_J = 125°C at I_{D2}		
	V dc	V dc Min	V dc Max	µA dc	Ω	Ω	°C/W	A (pk)
2N7454U1	-100	-2.0	-6.0	25	0.053	0.083	0.60	141
2N7455U1	-200				0.130	0.224		84

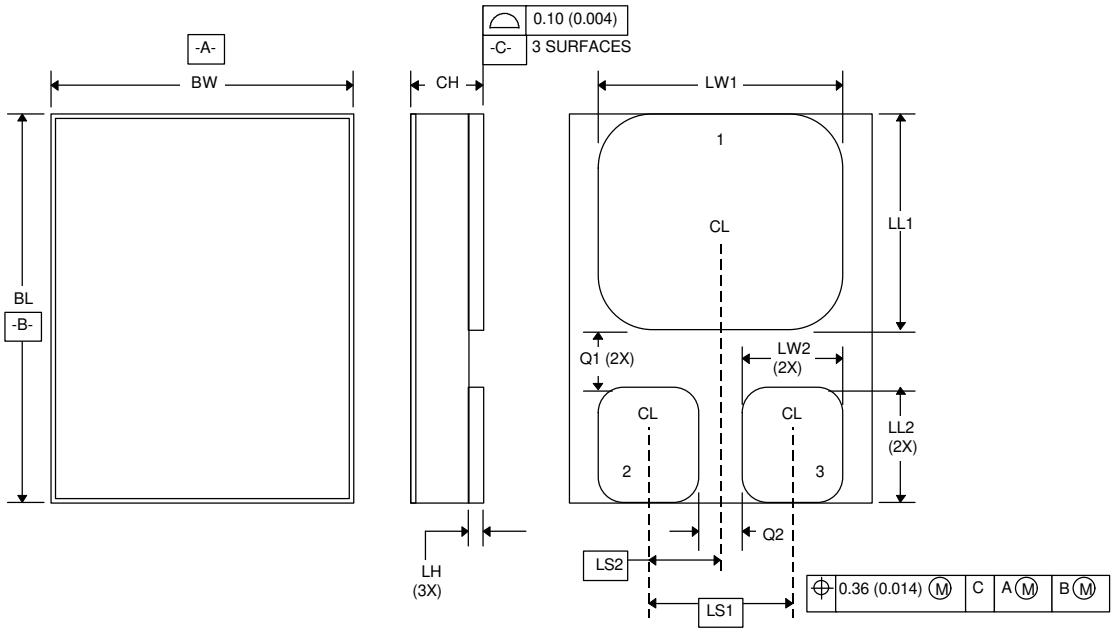
1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	17.38	17.67	0.684	0.696
BW	13.19	13.48	0.519	0.531
CH	---	3.60	---	0.142
LH	0.26	0.50	0.010	0.020
LW1	11.03	11.32	0.434	0.446
LW2	3.41	3.70	0.134	0.146
LL1	11.92	12.21	0.469	0.481
LL2	3.84	4.14	0.151	0.163
LS1	6.10 BSC		0.240 BSC	
LS2	3.05 BSC		0.120 BSC	
Q1	0.89	---	0.035	---
Q2	1.27	---	0.050	---
TERM 1	Drain			
TERM 2	Gate			
TERM 3	Source			

Notes:
1. Controlling dimension: Inch
2. Metric equivalents are given for information only.
3. The lid shall be electrically isolated from the Drain, Gate and Source.
4. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Dimensions and configuration (SMD-2).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, Building 4D (NPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

I_{AS} - Rated avalanche current, nonrepetitive

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and herein. Terminal material shall be copper-tungsten. Methods used for electrical isolation of the terminals shall employ materials that contain a minimum of 90 percent AL_2O_3 (ceramic).

3.3.1 Terminal finish. Terminal finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of terminal finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.7 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection.

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3.7.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended.

a. Devices should be handled on benches with conductive handling devices.
b. Ground test equipment, tools and personnel handling devices.
c. Do not handle devices by the leads.
d. Store devices in conductive foam or carriers.
e. Avoid use of plastic, rubber or silk in MOS areas.
f. Maintain relative humidity above 50 percent if practical.
g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.
h. Gate must be terminated to source, R ≤ or 100 kΩ, whenever bias voltage is applied drain to source.

3.8 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500, and table III herein. End point electrical measurements shall be as specified in table III herein.

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4.3 Screening (JANS only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement
	JANS
1/	Method 3470 E_{AS} test (see 4.5.4)
1/	Method 3161 (see 4.5.3)
1/	Gate stress test (see 4.5.5)
2/	Subgroup 2 of table I herein
9	I_{DSS1} , I_{GSS} as a minimum
10	MIL-STD-750, method 1042 test condition B
11	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(ON)1}$, $V_{GS(TH)}$ Subgroup 2 of table I herein ΔI_{GSSF1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100% of initial value, whichever is greater.
12	MIL-STD-750, method 1042 test condition A
13	Subgroups 2 and 3 of table I herein ΔI_{GSSF1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100% of initial value, whichever is greater. ΔI_{DSS1} = ±25µA dc or ±100% of initial value, whichever is greater. $\Delta r_{DS(ON)1}$ = ±20% of initial value. $\Delta V_{GS(TH)1}$ = ±20% of initial value.

1/ Shall be performed anytime before screen 10.
2/ Shall be performed after E_{AS} test, method 3161, and gate stress test.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables VIa (JANS) of MIL-PRF-19500 and paragraph 4.4.2.1 herein. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein. Delta V_{SD} measurements shall be in accordance with table IV herein.

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4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1051	Test condition G, 100 cycles
B3	2077	SEM
B4	1042	Intermittent operation life, condition D, 2,000 cycles. No heat sink nor or forced-air cooling on the device shall be permitted during the on cycle. T_{on} = 30 seconds minimum.
B5	1042	Accelerated steady-state reverse bias, condition A, V_{DS} = rated; T_A = +175°C; t = 120 hours, min.
B5	1042	Accelerated steady-state gate bias, condition B, V_{GS} = rated; T_A = +175°C; t = 24 hours.
B6	3161	Thermal resistance, see 4.5.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein. Delta V_{SD} measurements shall be in accordance with table IV herein.

Subgroup	Method	Condition
C2	2036	Not applicable.
C6	1042	Test condition D, 6,000 cycles; 1 cycle = 30 sec. min. No heat sink nor or forced-air cooling on the device shall be permitted during the on cycle. T_{on} = 30 seconds minimum.

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table VIII of MIL-PRF-19500 and table II herein.

4.4.4.1 Design parameters. Not tested on a per lot basis. Design shall be such that the devices shall be capable of meeting the requirements in figure 4.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal Resistance. Thermal Resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\theta JC}$ = 0.60 °C/W. The following parameters shall apply:

a. Measuring current(I_M)..... 10 mA
b. Drain heating current (I_H)..... 4 A
c. Heating time (t_H)..... Steady-state (see MIL-STD-750, method 3161)
d. Drain-source heating voltage (V_H)..... -20 V
e. Measurement time delay (t_{MD})..... 30 to 60 µs
f. Sample window time (t_{SW})..... 10 µs maximum

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TABLE I. Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Breakdown voltage drain to source	3407	V_{GS} = 0V, I_D = 1 mA dc, bias condition C	$V_{(BR)DSS}$			
2N7454U1				-100		V dc
2N7455U1				-200		V dc
Gate to source voltage (threshold)	3403	V_{DS} = V_{GS}, I_D = 1 mA dc	$V_{GS(TH)1}$	-2.0	-6.0	V dc
Gate current	3411	V_{GS} = ±20V dc, Bias condition C, V_{DS} = 0V	I_{GSS1}		±100	nA dc
Drain current	3413	V_{GS} = 0V dc, Bias condition C, V_{DS} = 80 % of rated V_{DS},	I_{DSS1}		25	µA dc
Static drain to source "ON" state resistance	3421	V_{GS} = -12V dc, condition A, pulsed (see 4.5.1), I_D = I_{D2}	$r_{DS(ON)1}$			
2N7454U1					0.053	Ω
2N7455U1					0.130	Ω
Static drain to source "ON" state voltage	3405	V_{GS} = -12V dc, condition A, pulsed (see 4.5.1), I_D = I_{D1}	$V_{DS(ON)1}$			
2N7454U1					-2.74	V dc
2N7455U1					-4.00	V dc
Forward voltage	4011	V_{GS} = 0V dc, condition A, pulsed (see 4.5.1), I_D = I_{D1}	V_{SD}		-1.8	V dc
Subgroup 3						
High temperature operation		T_C = T_J = +125°C				
Gate current	3411	V_{GS} = ±20V dc, Bias condition C, V_{DS} = 0V	I_{GSS2}		±200	nA dc
Drain current	3413	V_{GS} = 0V dc, Bias condition C, V_{DS} = 80% of rated V_{DS}	I_{DSS2}		0.25	mA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 3 – continued						
Static drain to source "ON"-state resistance	3421	V_{GS} = -12V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(ON)2}$			
2N7454U1					0.083	Ω
2N7455U1					0.224	Ω
Gate to source voltage (threshold)	3403	$V_{DS} = V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)2}$	-1.0		V dc
Low temperature operation		$T_C = T_J$ = -55°C				
Gate to source voltage (threshold)	3403	$V_{DS} = V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)3}$		-7.0	V dc
Subgroup 4						
Switching time test	3472	$I_D = I_{D1}$, V_{GS} = -12 V dc, R_G =2.35 Ω, V_{DD} = 50% of rated V_{DS}				
Turn-on delay time			$t_{D(on)}$			
2N7454U1					50	ns
2N7455U1					50	ns
Rise Time			t_r			
2N7454U1					75	ns
2N7455U1					25	ns
Turn-off delay time			$t_{D(off)}$			
2N7454U1					100	ns
2N7455U1					120	ns
Fall time			t_f			
2N7454U1					50	ns
2N7455U1					30	ns
Subgroup 5						
Safe operating area test (high voltage)	3474	See figure 3, t_p = 10 ms min. V_{DS} = 80% of max. rated V_{DS} ($V_{DS} \leq 200$ V)				
Electrical measurements		Table I, group A, subgroup 2 herein.				
Subgroup 6						
Not applicable						

See footnotes at end of table.

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TABLE I. Group A inspection - Continued

Inspection 1/	MIL-STD-750		Symbol	Limits		Units
	Method	Conditions		Min	Max	
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition A or B				
On-state gate charge			$Q_{G(ON)}$			
2N7454U1					250	nC
2N7455U1					250	nC
Gate to source charge			Q_{GS}			
2N7454U1					66	nC
2N7455U1					50	nC
Gate to drain charge			Q_{GD}			
2N7454U1					120	nC
2N7455U1					100	nC
Reverse recovery time	3473	di/dt = 100 A/µs, $V_{DD} \leq$ -50 V, $I_D = I_{D1}$	t_{rr}			
2N7454U1					200	ns
2N7455U1					310	ns

1/ For sampling plan, see MIL-PRF-19500.

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Table II. Group D Inspection

Inspection 1/, 2/, 3/, 4/	MIL-STD-750		Symbol	Pre-irradiation limits		Post irradiation limits		Units
	Method	Conditions		Min.	Max.	Min.	Max.	
Subgroup 1								
Not Applicable								
Subgroup 2		T_C = +25°C						
Steady state total dose irradiation (V_{GS} bias)	1019	V_{GS} = -12V, V_{DS} = 0V						
Steady state total dose irradiation (V_{DS} bias)	1019	V_{GS} = 0V, V_{DS} = 80% of rated V_{DS}						
Breakdown voltage drain to source	3407	V_{GS} = 0V, I_D = 1 mA dc, Bias condition C	$V_{(BR)DSS}$					
2N7454U1				-100		-100		
2N7455U1				-200		-200		
Gate to source Voltage (threshold)	3403	V_{DS} = V_{GS}, I_D = 1 mA dc	$V_{GS(TH)1}$	-2.0	-6.0	-2.0	-6.0	V dc
Gate current	3411	V_{GS} = ±20V dc, V_{DS} = 0V, Bias condition C	I_{GSS1}		±100		±100	nA dc
Drain current	3413	V_{GS} = 0V, V_{DS} = 80% of rated V_{DS}, bias condition C	I_{DSS1}		25		25	µA dc
Static drain to source "ON"-state resistance	3421	V_{GS} = -12V dc, condition A, pulsed (see4.5.1)						
2N7454U1		I_D = 28 A dc	$r_{DS(ON)3}$		0.055		0.055	Ω
2N7455U1		I_D = I_{D2}	$r_{DS(ON)1}$		0.130		0.130	Ω
Static drain to source "ON"-state voltage	3405	V_{GS} = -12V dc, condition A, Pulsed (see 4.5.1)						
2N7454U1		I_D = 44 A dc	$V_{DS(ON)2}$		-2.66		-2.66	V dc
2N7455U1		I_D = I_{D1}	$V_{DS(ON)1}$		-4.00		-4.00	V dc

1/ For sampling plan see MIL-PRF-19500.
2/ Electrical specifications are for 'D' and 'R' rad levels.
3/ Group D qualification may be performed prior to lot formation. Wafers qualified to these group D requirements may be used for any other detail specification utilizing the same die design.
4/ At the manufacturer's option, group D samples need not be subjected to all the screening tests, but shall be assembled in its qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.

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TABLE III. Group E inspection (all quality levels) - For qualification only

Inspection 1/, 2/, 3/, 4/, 5/	MIL-STD-750		Qualification and large lot quality conformance inspection.
	Method	Conditions	
Subgroup 1			12 devices c = 0
Temperature cycling (air to air)	1051	Test condition G, 500 cycles	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		Table I, group A, subgroup 2 herein.	
Subgroup 2 1/			12 devices c = 0
Steady-state gate bias	1042	Test condition B; 1,000 hours	
Electrical measurements		Table I, group A, subgroup 2 herein.	
Steady state reverse bias	1042	Test condition A; 1,000 hours	
Electrical measurements		Table I, group A, subgroup 2 herein.	
Subgroup 3			
Not applicable			
Subgroup 4			22 devices c = 0
Thermal resistance	3161	$R_{\theta JC}$ = 0.60°C/W maximum. See 4.5.2	
Subgroup 5			15 devices c = 0
Barometric pressure test (not required for $V_{BR(DSS)} \leq 200V$)	1001	Test condition C	

1/ See footnotes at end of table.

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TABLE III. Group E inspection (all quality levels) - For qualification only - continued.

Inspection 1/, 2/, 3/, 4/, 5/	MIL-STD-750		Qualification and large lot conformance inspection.
	Method	Conditions	
Subgroup 6			3 devices c = 0
ESD (Electrostatic Discharge Protection)	1020		
Subgroup 7			
Not applicable			
Subgroup 8			
Electrical measurements 3/		I_{GSS1} and I_{DSS1} tests from table I, group A, subgroup 2, herein.	3 devices 5/ c = 0
SEE Effect Testing 4/	1080	Fluence = 3e5 ±20% ions/cm^2 Flux = 5e3 to 2e4 ions/cm^{2}sec Beam energy = 260 to 300 MeV Temperature = 25°C ± 5°C	
2N7454U1		LET = 26 to 30 MeV-cm^2/mg Ion range = 40 to 45 microns Insitu bias conditions: V_{DS} = -100V and V_{GS} = 20V	
		LET = 36 to 40 MeV-cm^2/mg Ion range = 35 to 40 microns Insitu bias conditions: V_{DS} = -100V and V_{GS} = 10V V_{DS} = -80V and V_{GS} = 15V V_{DS} = -50V and V_{GS} = 20V	
2N7455U1		LET = 26 to 30 MeV-cm^2/mg Ion range = 40 to 45 microns Insitu bias conditions: V_{DS} = -200V and V_{GS} = 20V	
		LET = 36 to 40 MeV-cm^2/mg Ion range = 35 to 40 microns Insitu bias conditions: V_{DS} = -200V and V_{GS} = 5V V_{DS} = -160V and V_{GS} = 10V V_{DS} = -100V and V_{GS} = 15V V_{DS} = -40V and V_{GS} = 20V	
Electrical measurements 3/		I_{GSS1} and I_{DSS1} tests from table I, group A, subgroup 2, herein.	

1/ A separate sample for each test may be pulled.
2/ Group E qualification of single event effect testing may be performed prior to lot formation. Wafers qualified to these group E QCI requirements may be used for any other performance specification utilizing the same die design.
3/ As a minimum, gate to source leakages and drain to source leakage are to be examined to verify the electrical performance of the DUT prior to and after test. At the manufacturer's option, the remaining static tests in table I, group A, subgroup 2 may be performed.
4/ Devices passing a given combination of drain and gate voltage for an LET of 36 to 40 MeV-cm^2/mg qualify the same conditions for an LET of 26 to 30 MeV-cm^2/mg.
5/ This sampling plan applies to each bias condition defined.

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TABLE IV. Groups B and C delta V_{SD} measurements

Step	Inspection 1/, 2/	MIL-STD-750		Symbol	Limits		Units
		Method	Conditions		Min	Max	
1.	Thermal response	3161	See 4.5.3	ΔV_{SD}		55	mV

1/ The delta V_{SD} measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 4, see table IV herein, step 1.

2/ The delta V_{SD} measurements for table VII of MIL-PRF-19500 are as follows:

a. Subgroup 6, see table IV herein, steps 1.

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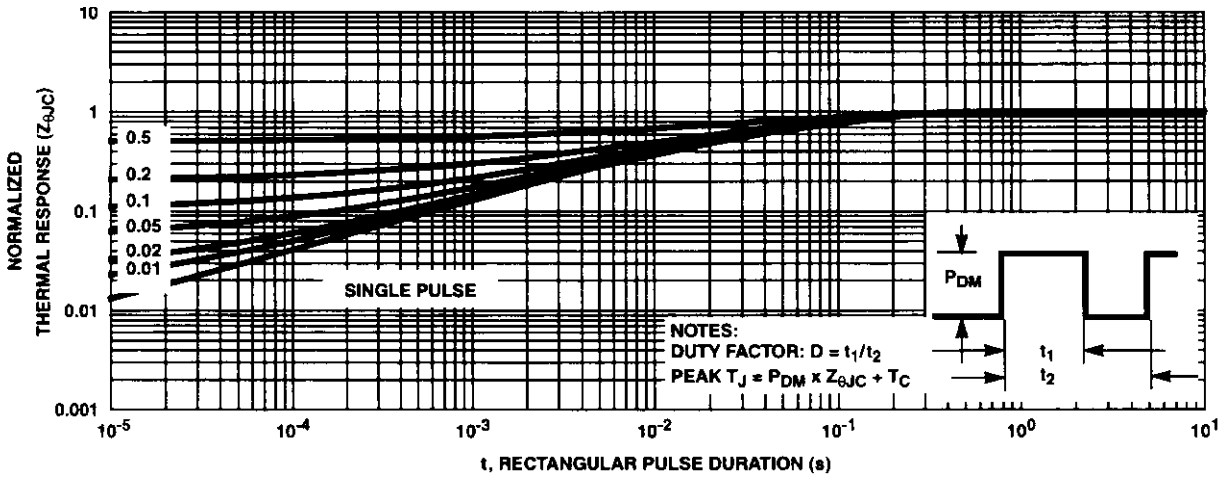


FIGURE 2. Thermal response curves

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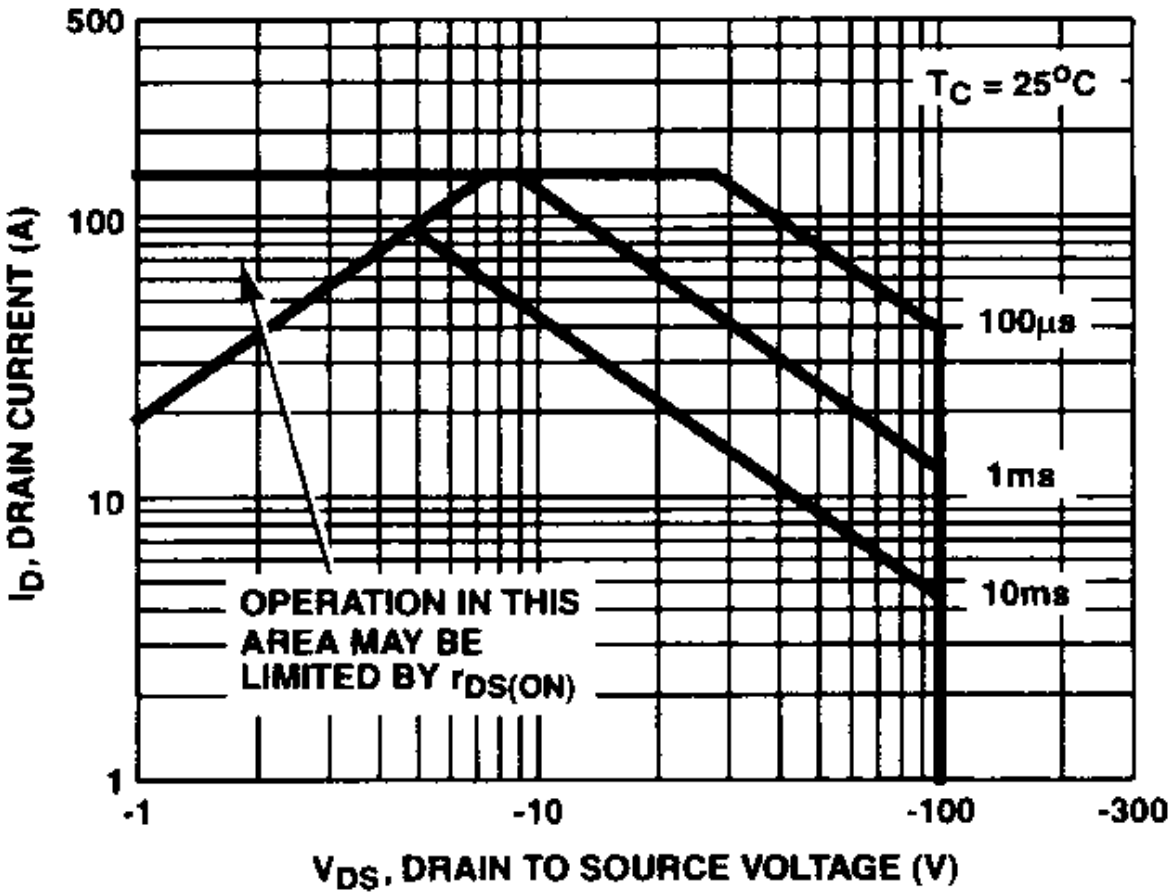


FIGURE 3. Safe operating area graphs.

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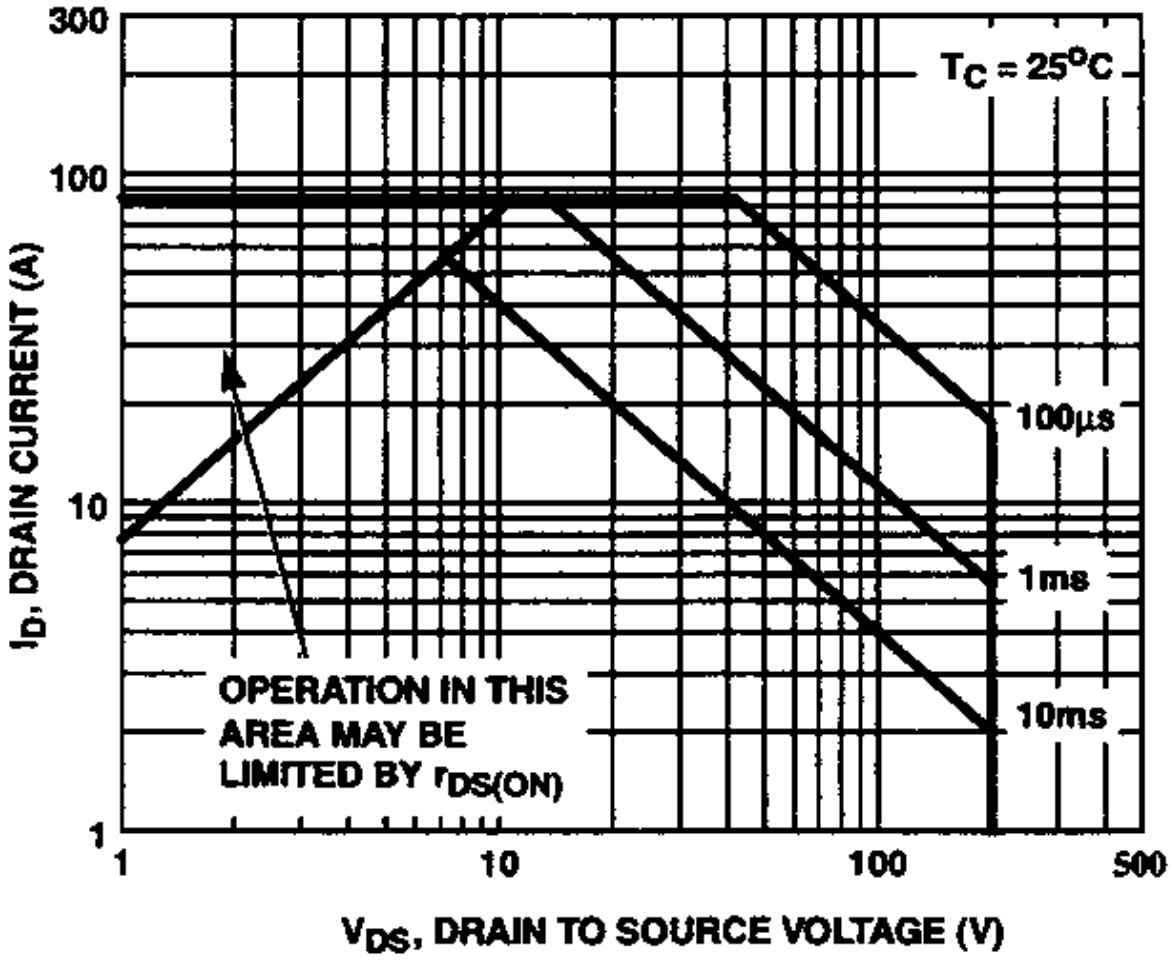


FIGURE 3. Safe operating area graphs - continued.

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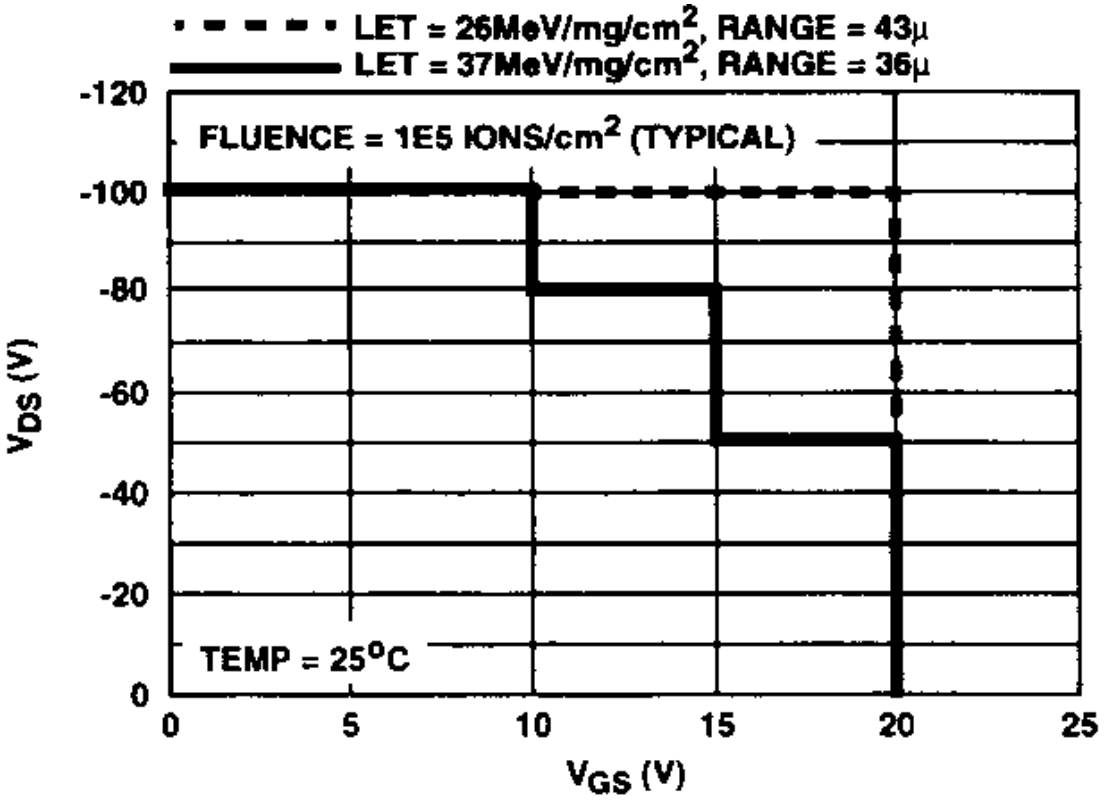


FIGURE 4. Single event effects safe operating area graphs.

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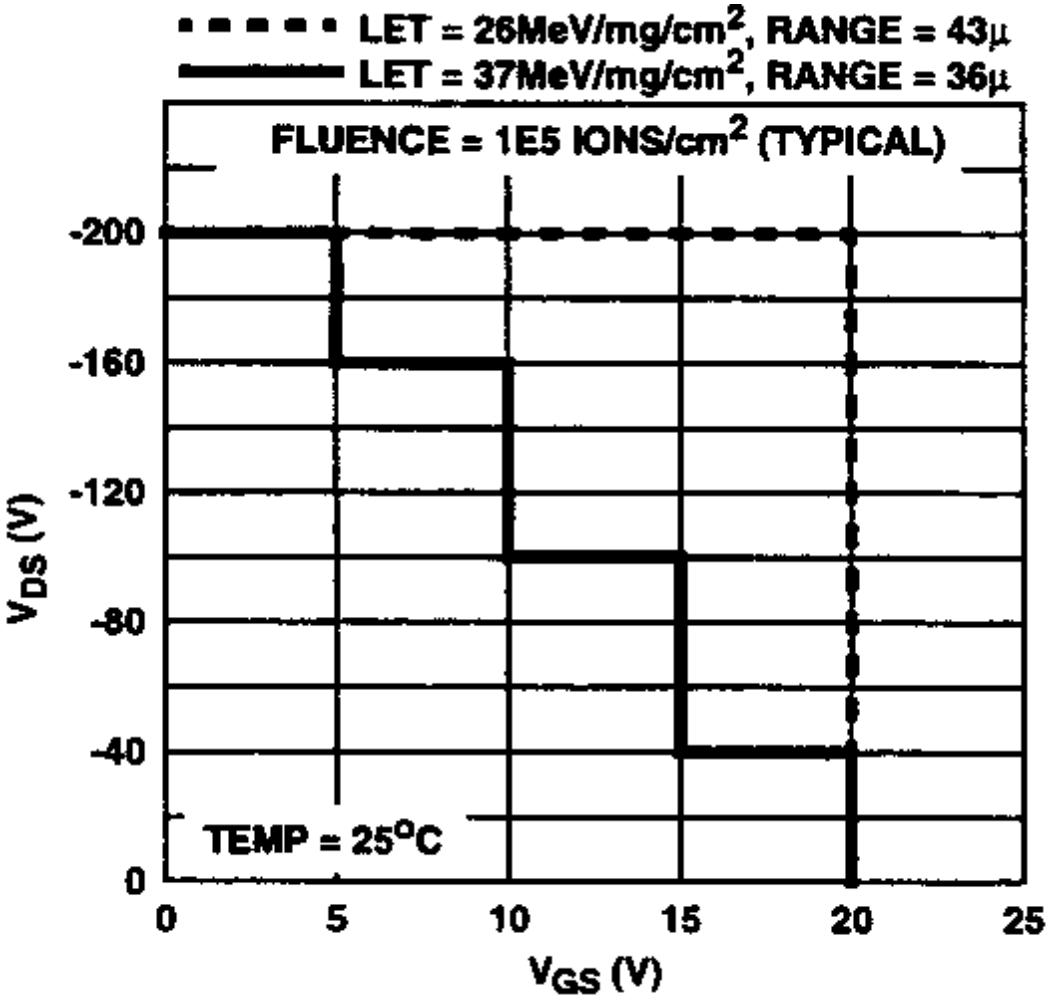


FIGURE 4. Single event effects safe operating area graphs - continued.

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4.5.3 Thermal Response ΔV_{SD} measurement). The delta V_{SD} measurement shall be performed in accordance with method 3161 of MIL-STD-750. The delta V_{SD} conditions (I_H and V_H) and maximum limit shall be derived by each vendor from the thermal response curves (see figure 2) and shall be specified in the certificate of conformance prior to qualification. The following parameter measurements shall apply:

a. Measuring current (I_M)..... 10 mA

b. Drain heating current (I_H)..... 4 A

c. Heating time (t_H)..... 10 ms

d. Drain-source heating voltage (V_H)..... -25 V

e. Measurement time delay (t_{MD})..... 30 - 60 µs

f. Sample window time (t_{SW})..... 10 µs maximum

4.5.4 Single pulse avalanche energy (E_{AS}).

a. $I_{AS} = I_{DM}$

b. L = 0.1 mH

c. Gate to source resistor ($25 \leq R_{GS} \leq 200\Omega$)

d. $E_{AS} = 1/2\ LI_{AS}^2$

e. VDD = -50 V to -150 V dc

f. Initial junction temperature = 25°C, -5°C, +10°C

4.5.5 Gate stress test.

a. V_{GS} = -30 V

b. t = 250 µs, minimum

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1).

b. Lead finish (see 3.3.1).

c. Type designation and product assurance level.

d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

6.4 Cross-reference list. The following table shows the generic P/N and its associated military P/N (without JAN and RHA prefix).

Generic P/N	Military P/N JANSD, R
SYC9160 SYC9260	2N7454U1 2N7455U1

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2121)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/666	**2. DOCUMENT DATE (YYMMDD)** 99/08/03

DOCUMENT TITLE SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED (TOTAL DOSE AND SINGLE EVENT EFFECTS) TRANSISTOR, P-CHANNEL SILICON TYPE 2N7454U1, 2N7455U1 JANSD, R

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. Point of contact: Alan Barone	b. TELEPHONE Commercial: 614-692-0510 DSN: 850-0510 FAX: 614-692-6939 EMAIL: alan_barone@dscc.dla.mil
c. ADDRESS: Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

DD FORM 1426, FEB 99 (EG) PREVIOUS EDITIONS ARE OBSOLETE WHS/DIOR, Feb 99

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